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Duration	29 mins 47 secs
Grade	**26.00** out of 40.00 (**65**%)

Question 1

Incorrect

Mark 0.00 out of 1.00

Which of the following is a scalar?

Select one:

- a. 5
- b. A 2×2 matrix
- c. (2, 3) Incorrect option.
- d. [1, 4, 7]

Incorrect.

The correct answer is: 5

Question 2

Incorrect

Mark 0.00 out of 1.00

Vector projection of vector a onto vector b gives:

Select one:

- a. The component of a in the direction of b
- b. A scalar equal to |a||b| Incorrect option.
- c. A vector perpendicular to b
- d. The sum of a and b

Incorrect.

The correct answer is: The component of a in the direction of b

Question 3

Correct

Mark 1.00 out of 1.00

A vector space must be closed under:

Select one:

- a. Addition only
- b. Addition and scalar multiplication ✓ Correct answer.
- c. Scalar multiplication only
- d. Matrix multiplication only

Correct.

The correct answer is: Addition and scalar multiplication

Question 4

Correct

Mark 1.00 out of 1.00

A linear combination of vectors v1 and v2 has the form:

Select one:

- a. v1 × v2 only
- b. c1v1 + c2v2 ✓ Correct answer.
- c. v1 / v2
- d. |v1| + |v2|

Correct.

The correct answer is: c1v1 + c2v2

Question 5

Correct

Mark 1.00 out of 1.00

The span of a set of vectors is:

Select one:

- a. The set of all linear combinations of those vectors — Correct answer.
- b. The number of vectors in the set
- c. The norm of the longest vector
- d. Only the zero vector

Correct.

The correct answer is: The set of all linear combinations of those vectors

Question 6

Correct

Mark 1.00 out of 1.00

Vectors are linearly independent if:

Select one:

- a. They all have the same length
- b. Their dot products are always 1
- c. No vector can be written as a linear combination of the others — Correct answer.
- d. They are pairwise orthogonal only

Correct.

The correct answer is: No vector can be written as a linear combination of the others

Question 7

Incorrect

Mark 0.00 out of 1.00

Which set is linearly dependent?

Select one:

- a. (1, 2), (2, 4)
- b. (1, 0, 0), (0, 1, 0), (0, 0, 1)
- c. (1, 1), (1, -1)
- d. (1, 0), (0, 1) Incorrect option.

Incorrect.

The correct answer is: (1, 2), (2, 4)

Question 8

Correct

Mark 1.00 out of 1.00

A basis of a vector space is a set of vectors that is:

Select one:

- a. Square in shape
- b. Minimal in norm
- c. Linearly independent and spanning Correct answer.
- d. Orthogonal and positive

Correct.

The correct answer is: Linearly independent and spanning

Question 9

Correct

Mark 1.00 out of 1.00

In R^3, the maximum number of linearly independent vectors is:

Select one:

- a. 3 Correct answer.
- b. 2
- c. 4
- d. 1

Correct.

The correct answer is: 3

Question 10

Correct

Mark 1.00 out of 1.00

Orthonormal vectors are vectors that are:

Select one:

- a. Always in R^2 only
- b. Independent but not normalized
- c. Orthogonal and each has unit norm — Correct answer.
- d. Parallel and same length

Correct.

The correct answer is: Orthogonal and each has unit norm

Question 11

Incorrect

Mark 0.00 out of 1.00

A linear transformation T must satisfy:

Select one:

- a. $T(x + y) = T(x) + T$ and $T(cx) = cT(x)$
- b. $T(cx) = T(x) + c$
- c. $T(x + y) = T(xy)$ Incorrect option.
- d. $T(0) \neq 0$

Incorrect.

The correct answer is: $T(x + y) = T(x) + T$ and $T(cx) = cT(x)$

Question 12

Correct

Mark 1.00 out of 1.00

If $u = (2, 1)$ and $v = (3, 4)$, then $u + v =$

Select one:

- a. $(5, 5)$ Correct answer.
- b. $(5, 4)$
- c. $(1, 3)$
- d. $(6, 4)$

Correct.

The correct answer is: $(5, 5)$

Question 13

Incorrect

Mark 0.00 out of 1.00

Which of the following is generally not a linear transformation?

Select one:

- a. $T(x, y) = (2x, 2y)$
- b. $T(x, y) = (x + 1, y)$
- c. $T(x, y) = (0, 0)$ Incorrect option.
- d. $T(x, y) = (x + y, y)$

Incorrect.

The correct answer is: $T(x, y) = (x + 1, y)$

Question 14

Correct

Mark 1.00 out of 1.00

An affine space differs from a vector space because it:

Select one:

- a. Includes translation and need not pass through the origin — Correct answer.
- b. Has no dimensions
- c. Contains only orthogonal vectors
- d. Does not allow addition

Correct.

The correct answer is: Includes translation and need not pass through the origin

Question 15

Correct

Mark 1.00 out of 1.00

The rank of a matrix is the dimension of its:

Select one:

- a. Diagonal
- b. Column space or row space — Correct answer.
- c. Determinant
- d. Null space only

Correct.

The correct answer is: Column space or row space

Question 16

Incorrect

Mark 0.00 out of 1.00

If A and B are matrices of the same size, then A + B is defined by:

Select one:

- a. Multiplying corresponding entries ⊗ Incorrect option.
- b. Multiplying rows only
- c. Adding diagonals only
- d. Adding corresponding entries

Incorrect.

The correct answer is: Adding corresponding entries

Question 17

Incorrect

Mark 0.00 out of 1.00

The transpose of a matrix is obtained by:

Select one:

- a. Replacing each entry by its inverse ⊗ Incorrect option.
- b. Swapping rows and columns
- c. Multiplying by identity
- d. Changing signs of all entries

Incorrect.

The correct answer is: Swapping rows and columns

Question 18

Incorrect

Mark 0.00 out of 1.00

If A is a 2×3 matrix and B is a 3×4 matrix, then AB has size:

Select one:

- a. 2×3
- b. 3×3 ⊗ Incorrect option.
- c. 2×4
- d. 4×2

Incorrect.

The correct answer is: 2×4

Question 19

Correct

Mark 1.00 out of 1.00

Matrix multiplication is generally:

Select one:

- a. Non-commutative ✓ Correct answer.
- b. Non-associative
- c. Undefined for square matrices
- d. Commutative

Correct.

The correct answer is: Non-commutative

Question 20

Correct

Mark 1.00 out of 1.00

The Hadamard product of two matrices means:

Select one:

- a. Transpose multiplication
- b. Standard matrix multiplication
- c. Entrywise multiplication ✓ Correct answer.
- d. Entrywise addition

Correct.

The correct answer is: Entrywise multiplication

Question 21

Correct

Mark 1.00 out of 1.00

The identity matrix I has the property that:

Select one:

- a. Its determinant is always 0
- b. AI = I for all A
- c. All entries are 1
- d. AI = A whenever dimensions are compatible ✓ Correct answer.

Correct.

The correct answer is: AI = A whenever dimensions are compatible

Question 22

Correct

Mark 1.00 out of 1.00

A square matrix is invertible if and only if:

Select one:

- a. Its determinant is nonzero ✓ Correct answer.
- b. It is symmetric
- c. Its determinant is 0
- d. All entries are positive

Correct.

The correct answer is: Its determinant is nonzero

Question 23

Correct

Mark 1.00 out of 1.00

If v = (2, -1, 3), then 2v =

Select one:

- a. (4, -1, 3)
- b. (4, -2, 6) ✓ Correct answer.
- c. (1, -2, 6)
- d. (2, -2, 6)

Correct.

The correct answer is: (4, -2, 6)

Question 24

Correct

Mark 1.00 out of 1.00

The determinant of a matrix gives information about:

Select one:

- a. Orthogonality only
- b. Scaling effect and invertibility — Correct answer.
- c. Only the trace
- d. Number of rows only

Correct.

The correct answer is: Scaling effect and invertibility

Question 25

Correct

Mark 1.00 out of 1.00

The trace of a square matrix is:

Select one:

- a. Product of diagonal entries
- b. Sum of all entries
- c. Rank of the matrix
- d. Sum of diagonal entries — Correct answer.

Correct.

The correct answer is: Sum of diagonal entries

Question 26

Correct

Mark 1.00 out of 1.00

A symmetric matrix satisfies:

Select one:

- a. A^-1 = A
- b. A^T = -A
- c. A^T = A ✓ Correct answer.
- d. det(A) = 1

Correct.

The correct answer is: A^T = A

Question 27

Incorrect

Mark 0.00 out of 1.00

An orthogonal matrix Q satisfies:

Select one:

- a. QQ = I for all Q
- b. Q + Q^T = 0
- c. det(Q) = 0 ✗ Incorrect option.
- d. Q^TQ = I

Incorrect.

The correct answer is: Q^TQ = I

Question 28

Incorrect

Mark 0.00 out of 1.00

A diagonal matrix is a matrix in which:

Select one:

- a. Every row is identical
- b. All diagonal entries are zero — Incorrect option.
- c. All non-diagonal entries are zero
- d. All entries are equal

Incorrect.

The correct answer is: All non-diagonal entries are zero

Question 29

Correct

Mark 1.00 out of 1.00

A positive definite matrix A satisfies:

Select one:

- a. A has all zero eigenvalues
- b. A is never symmetric
- c. x^T A x > 0 for all nonzero x — Correct answer.
- d. det(A) < 0 always

Correct.

The correct answer is: x^T A x > 0 for all nonzero x

Question 30

Correct

Mark 1.00 out of 1.00

A low-rank matrix approximation is mainly used for:

Select one:

- a. Changing vectors into scalars
- b. Compression and noise reduction — Correct answer.
- c. Making every matrix invertible
- d. Increasing matrix size

Correct.

The correct answer is: Compression and noise reduction

Question 31

Correct

Mark 1.00 out of 1.00

For a matrix A, a nonzero vector v is an eigenvector if:

Select one:

- a. $vA = \lambda$
- b. $A + v = \lambda$
- c. $Av = 0$
- d. $Av = \lambda v$ for some scalar λ — Correct answer.

Correct.

The correct answer is: $Av = \lambda v$ for some scalar λ

Question 32

Correct

Mark 1.00 out of 1.00

Eigenvalues are useful because they help describe:

Select one:

- a. Important directions and scaling behavior of transformations — Correct answer.
- b. Only matrix size
- c. Matrix addition rules only
- d. Row order only

Correct.

The correct answer is: Important directions and scaling behavior of transformations

Question 33

Correct

Mark 1.00 out of 1.00

Principal Component Analysis (PCA) is mainly used for:

Select one:

- a. Solving differential equations
- b. Sorting vectors alphabetically
- c. Matrix inversion
- d. Dimensionality reduction — Correct answer.

Correct.

The correct answer is: Dimensionality reduction

Question 34

Incorrect

Mark 0.00 out of 1.00

The dot product of (1, 2, 3) and (4, 0, -1) is

Select one:

- a. 1
- b. 2 — Incorrect option.
- c. 6
- d. 4

Incorrect.

The correct answer is: 1

Question 35

Incorrect

Mark 0.00 out of 1.00

In PCA, the principal components correspond to directions of:

Select one:

- a. Equal variance always
- b. Zero variance
- c. Minimum variance only — Incorrect option.
- d. Maximum variance

Incorrect.

The correct answer is: Maximum variance

Question 36

Correct

Mark 1.00 out of 1.00

Two vectors are orthogonal if their:

Select one:

- a. Norms are equal
- b. Dot product is zero — Correct answer.
- c. Sum is zero
- d. Dimensions are different

Correct.

The correct answer is: Dot product is zero

Question 37

Correct

Mark 1.00 out of 1.00

Cosine similarity between two vectors measures:

Select one:

- a. Sum of their coordinates
- b. Difference in their lengths
- c. Angle-based similarity between them — Correct answer.
- d. Product of their norms

Correct.

The correct answer is: Angle-based similarity between them

Question 38

Incorrect

Mark 0.00 out of 1.00

If two nonzero vectors are orthogonal, their cosine similarity is:

Select one:

- a. Undefined in all cases
- b. 0
- c. -1 ⊗ Incorrect option.
- d. 1

Incorrect.

The correct answer is: 0

Question 39

Incorrect

Mark 0.00 out of 1.00

The Euclidean norm of vector (3, 4) is

Select one:

- a. 5
- b. 6 ⊗ Incorrect option.
- c. 12
- d. 7

Incorrect.

The correct answer is: 5

Question 40

Correct

Mark 1.00 out of 1.00

A unit vector is a vector whose norm is:

Select one:

- a. Equal to its dimension
- b. 0
- c. 2
- d. 1 Correct answer.

Correct.

The correct answer is: 1

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Assignment 1